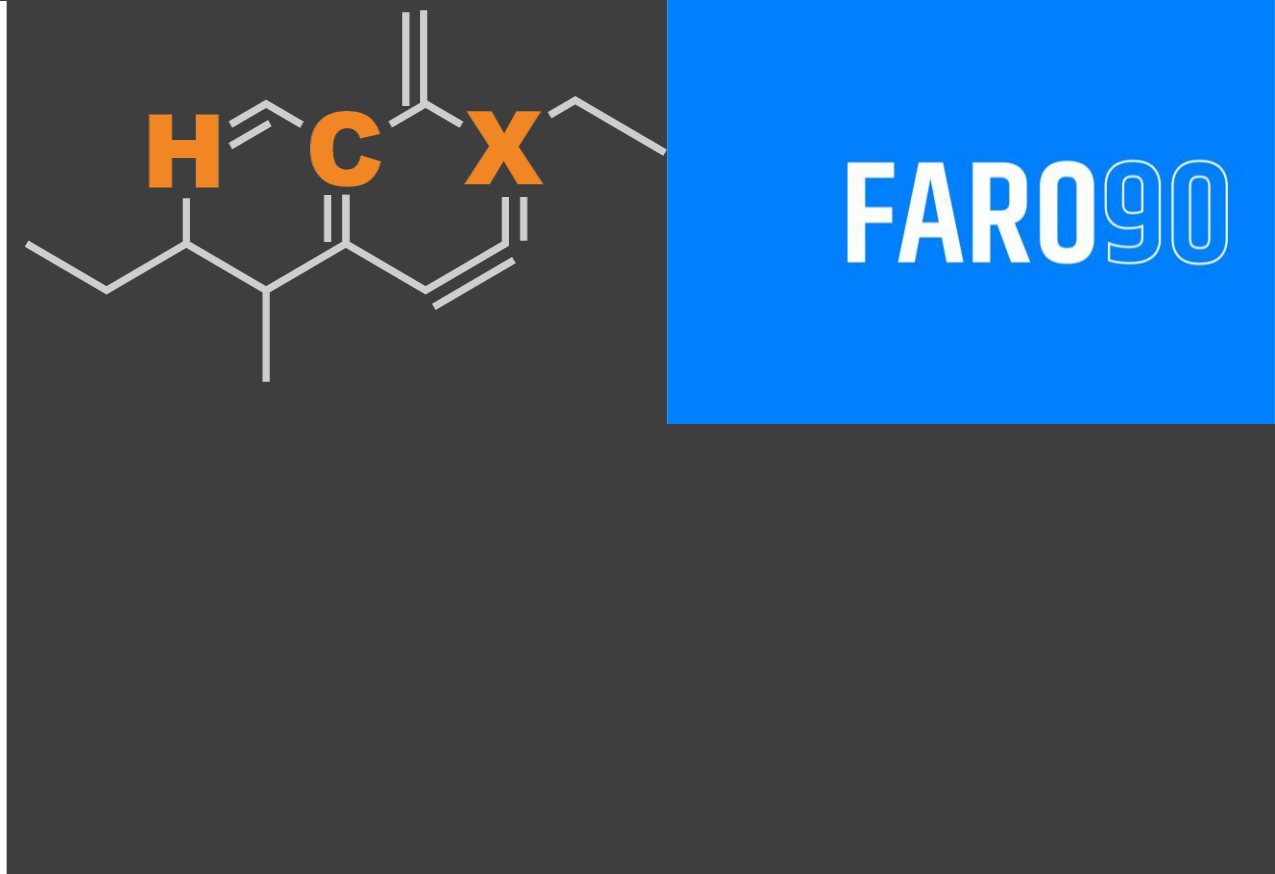




Ethanol Blending in Gasoline - Italy

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Italy

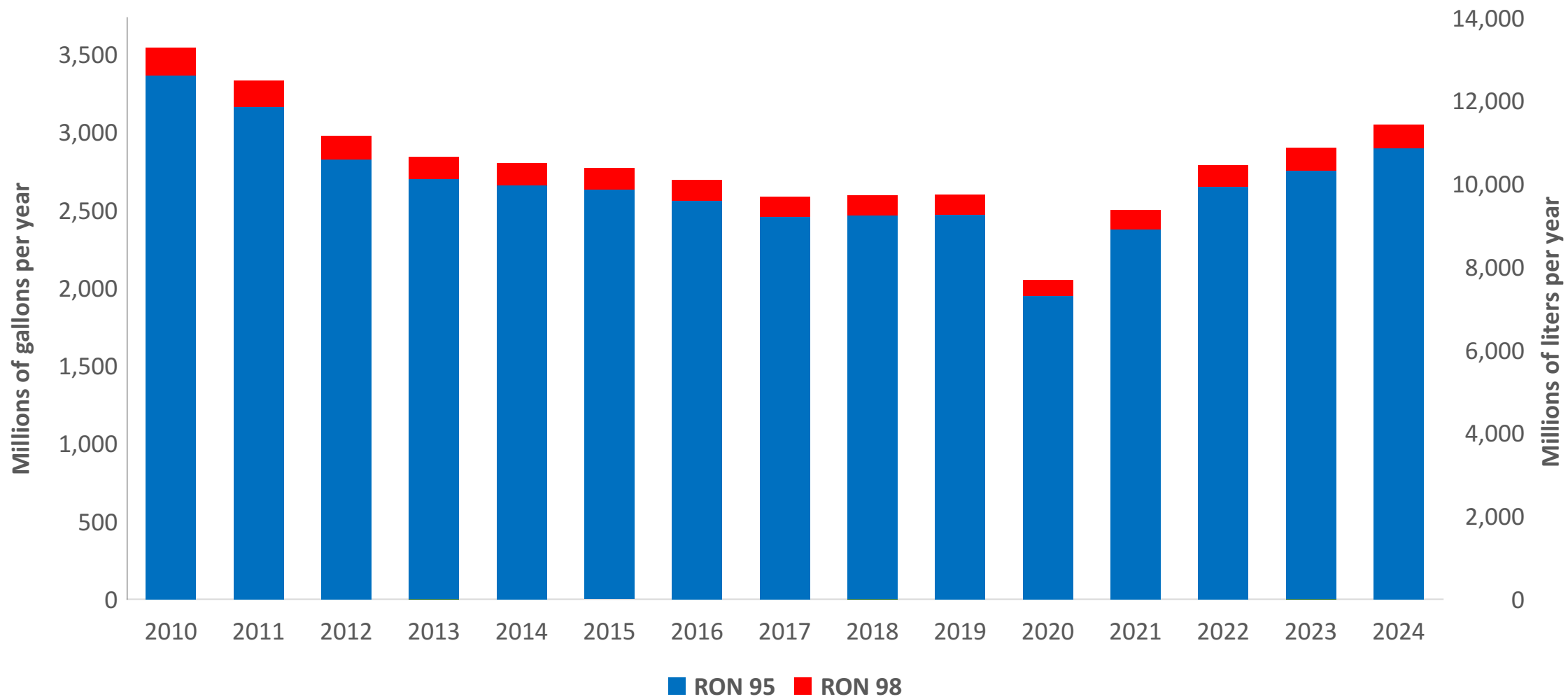


In 2024, gasoline consumption in Italy was 11,426 million liters (3,018 million gallons) and growth rate was 3.2%. Gasoline production and net imports were 17,947 and -6,521 million liters (4,741 and -1,723 million gallons) correspondingly. Italy complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Italy was 640 million liters (169 million gallons) representing 5.6% of gasoline demand. Ethanol production and Net Imports were 700 and -60 million liters (169 and -16 million gallons) correspondingly. Ethanol sources are Wine and Second Generation Lignocelulosic ethanol.

Gasoline Demand in Italy

Gasoline Demand by Grade in Italy

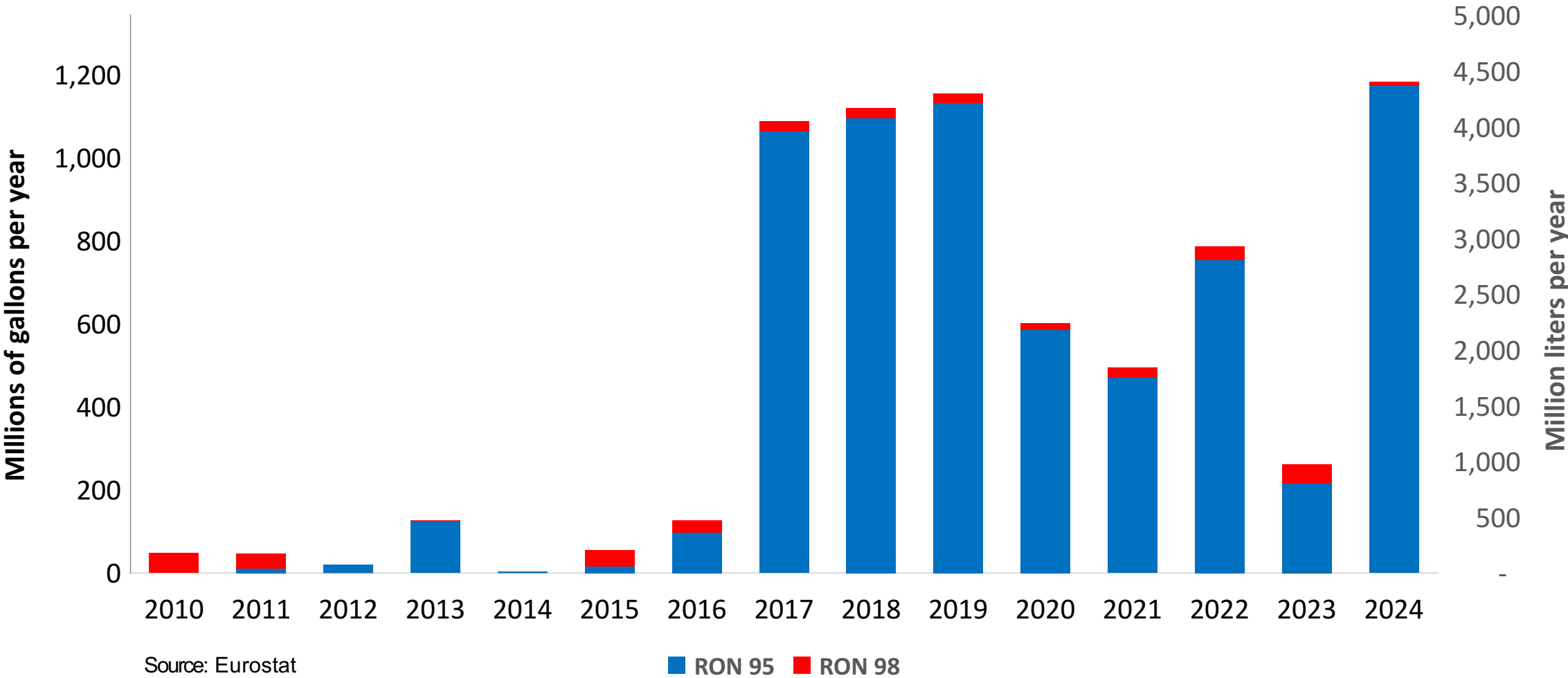


Source: Eurostat

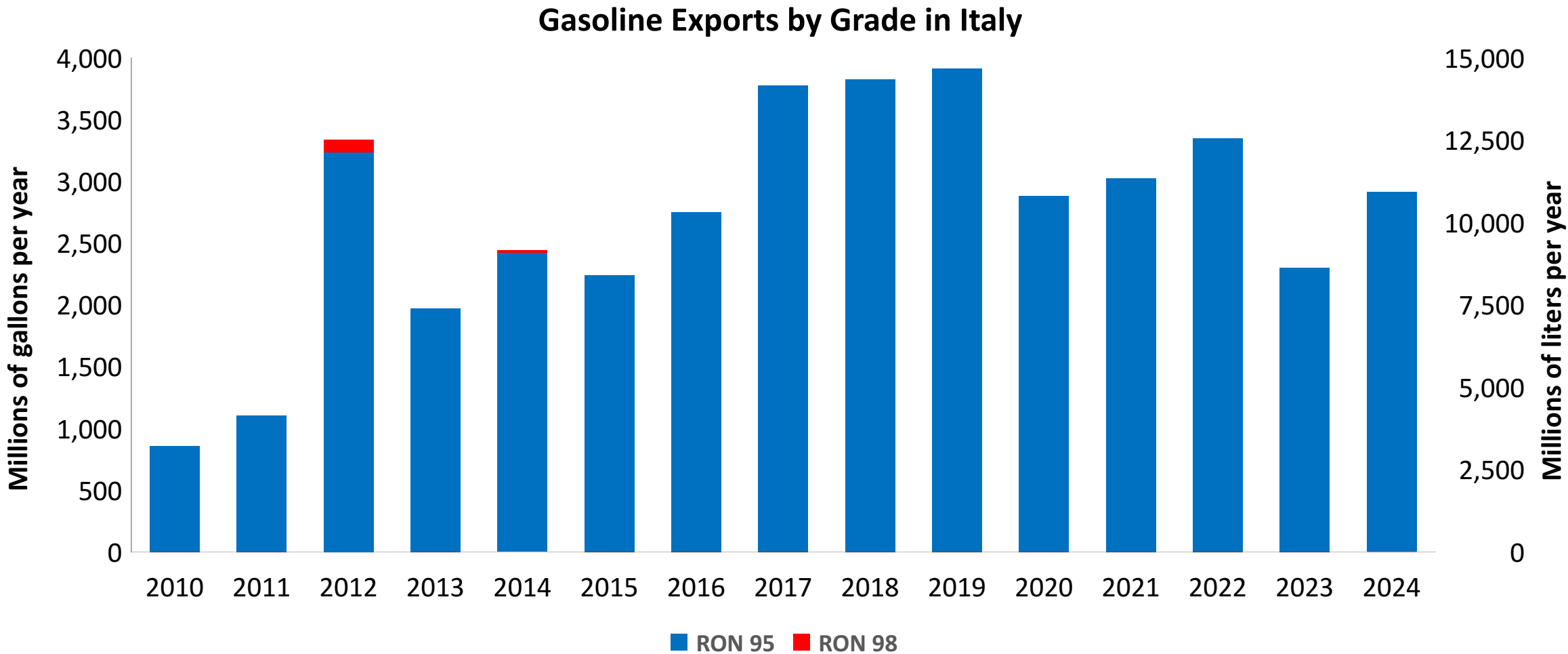
The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a cyclohexene ring with a methyl group at position 1, a vinyl group at position 2, and a substituent 'X' at position 3. The ring carbons are labeled with 'H' and 'C' in orange, and the substituent 'X' is also in orange.



Gasoline Imports by Grade in Italy



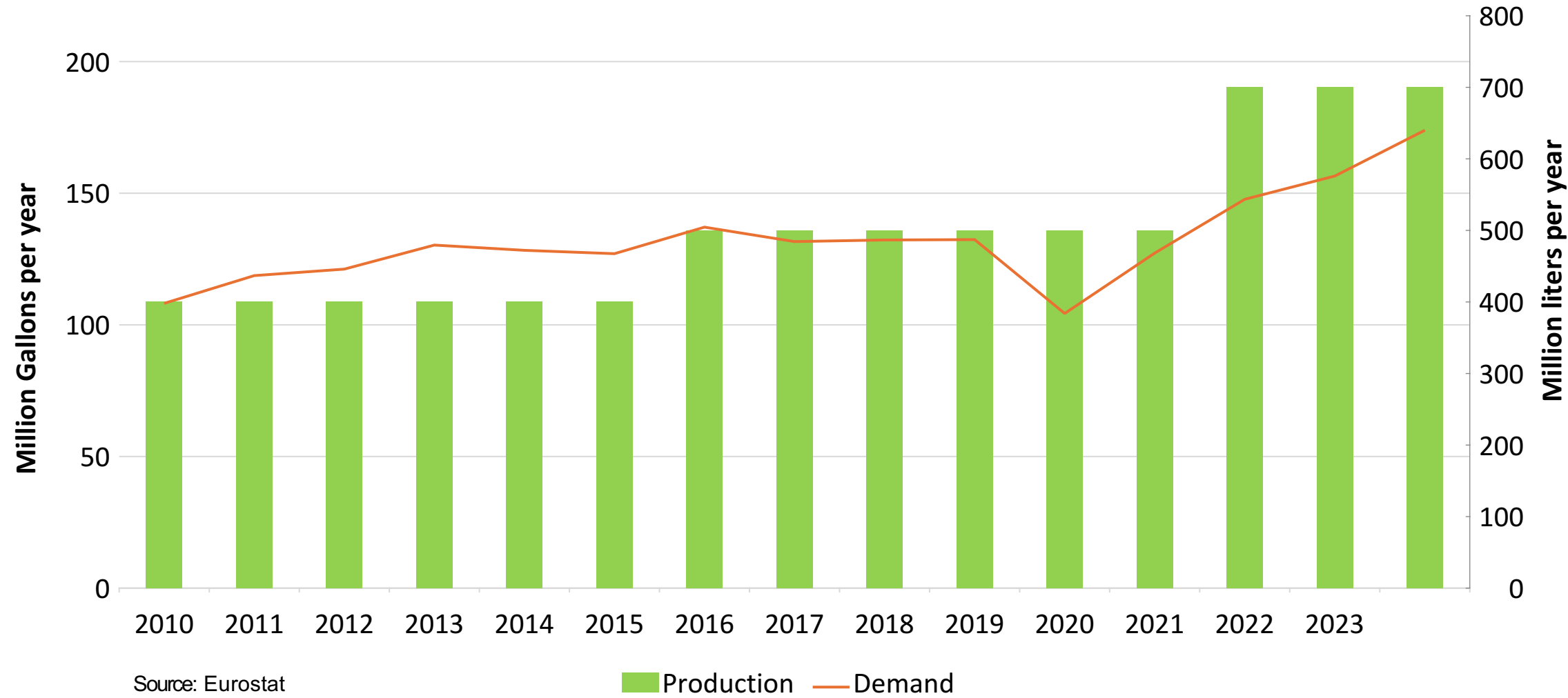
Gasoline Exports in Italy



Source: Eurostat

Ethanol Balance in Italy

Ethanol Balance in Italy



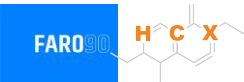
Gasoline Quality in Italy

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.7							
Advanced Biofuels (%cal)	4.9							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

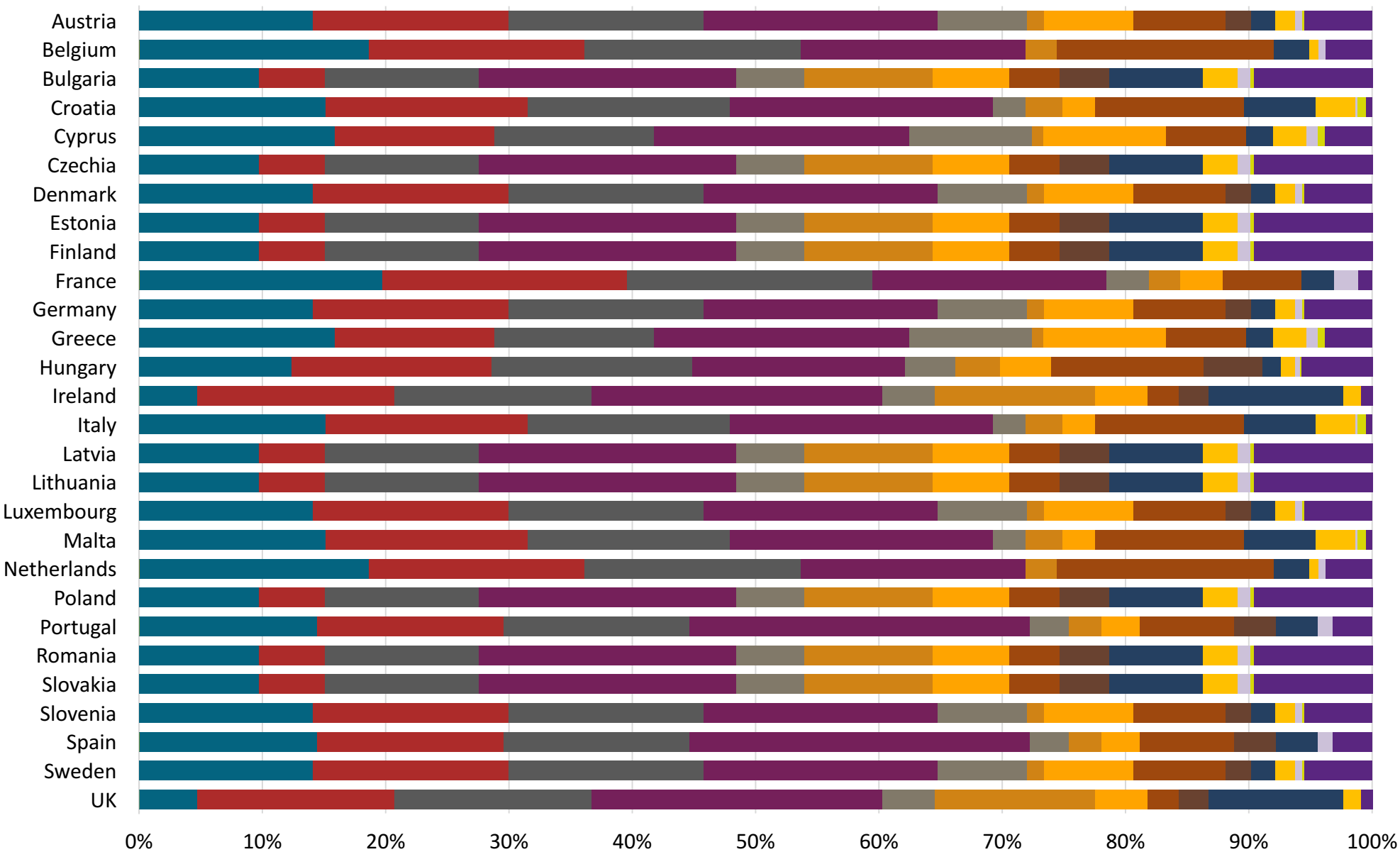
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



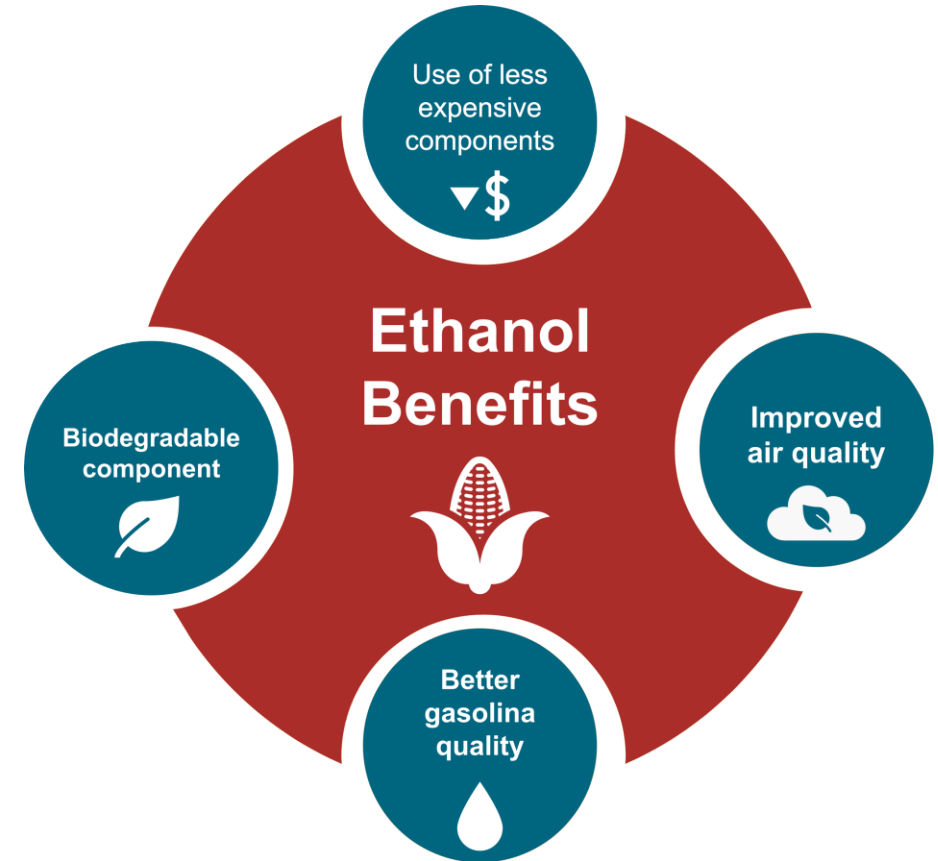
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

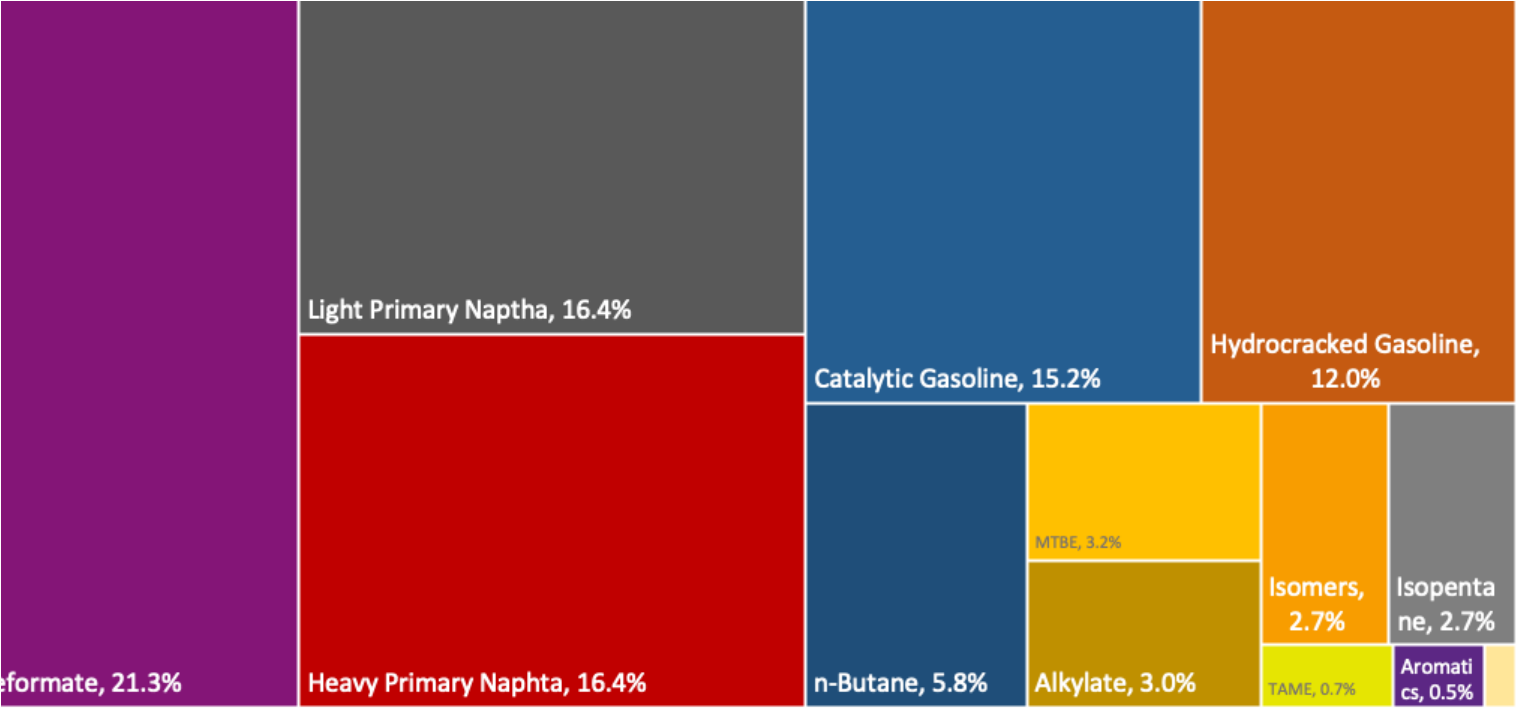
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

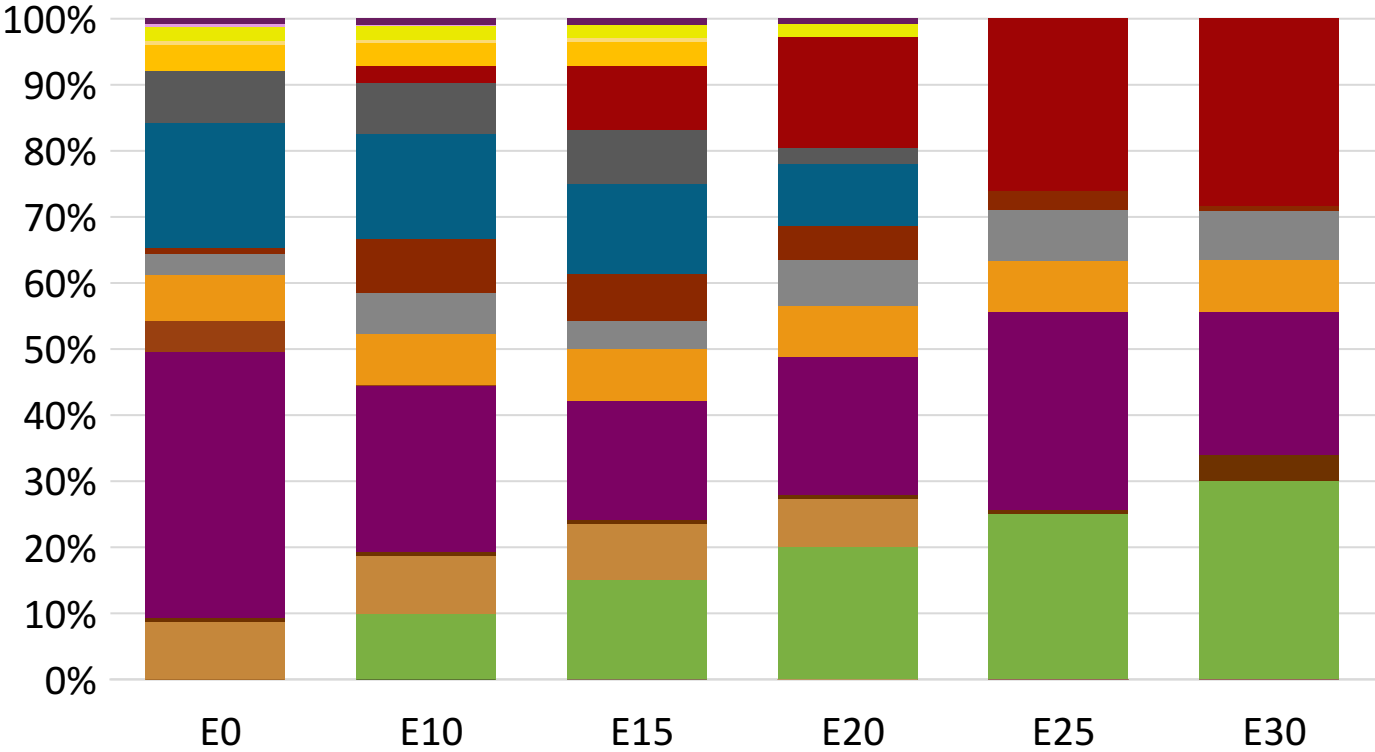


Italy Available Blending Components



ETBE,
0.2%

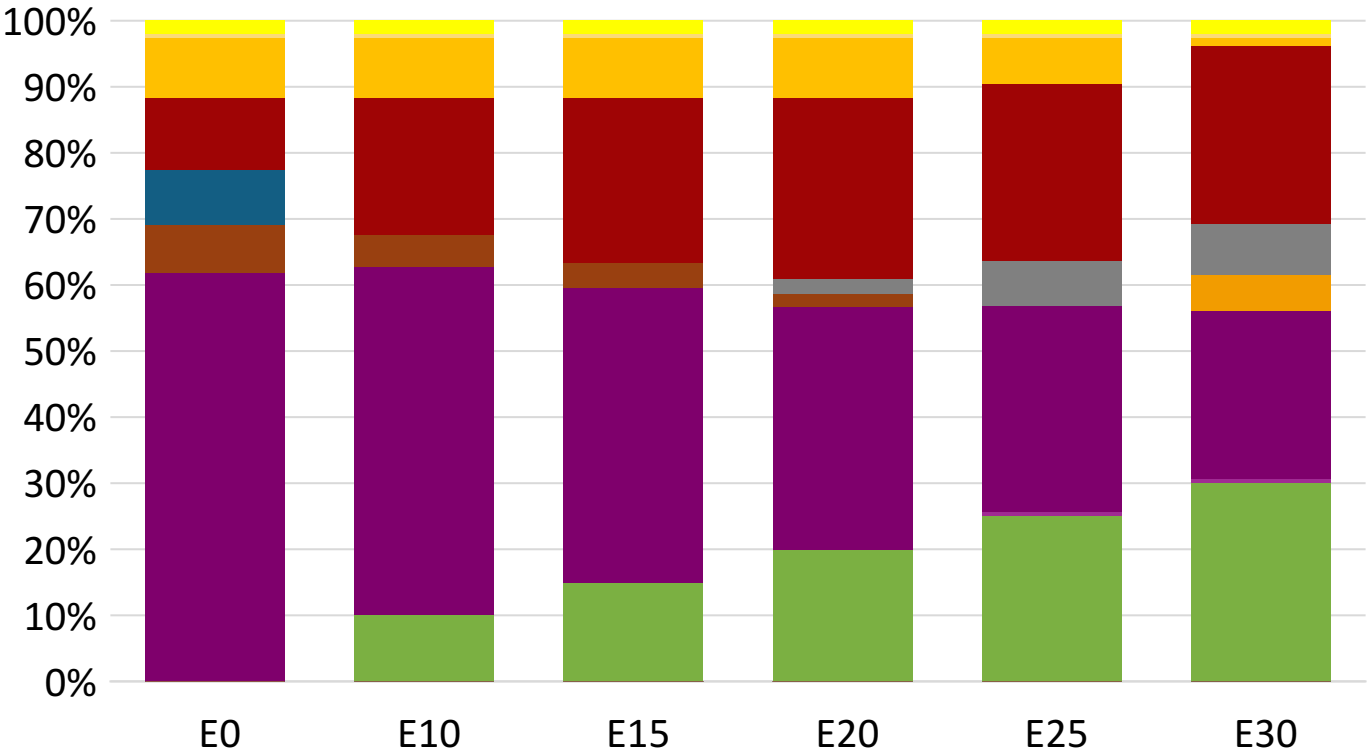
Italy – Gasoline 95 – Constant Octane



Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.258	\$ 2.145	\$ 2.062	\$ 1.986	\$ 1.891	\$ 1.817

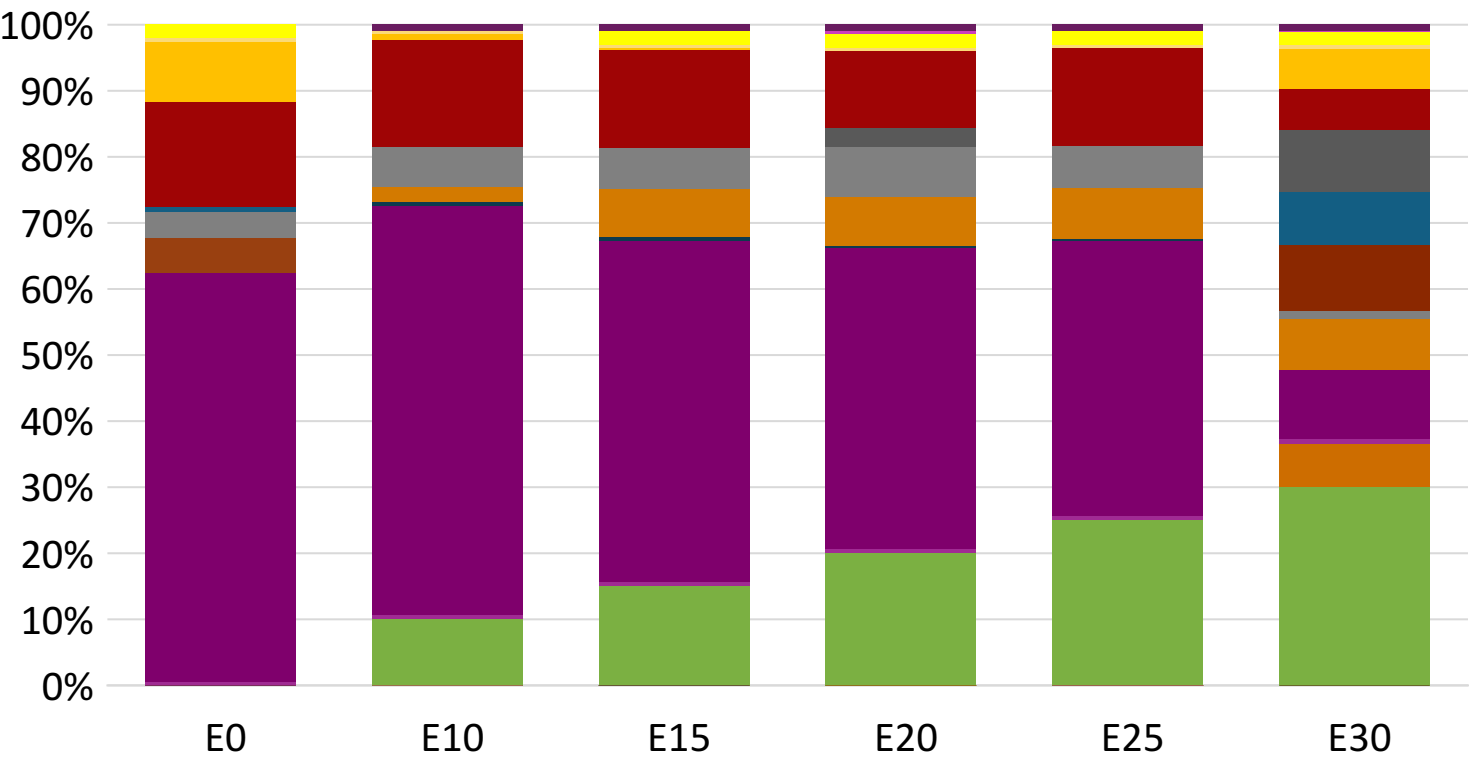
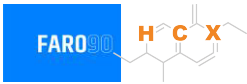
Italy - Gasoline 98 - Constant Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.307	\$ 2.188	\$ 2.119	\$ 2.049	\$ 1.979	\$ 1.926

Italy – Gasoline 95 – Increased Octane

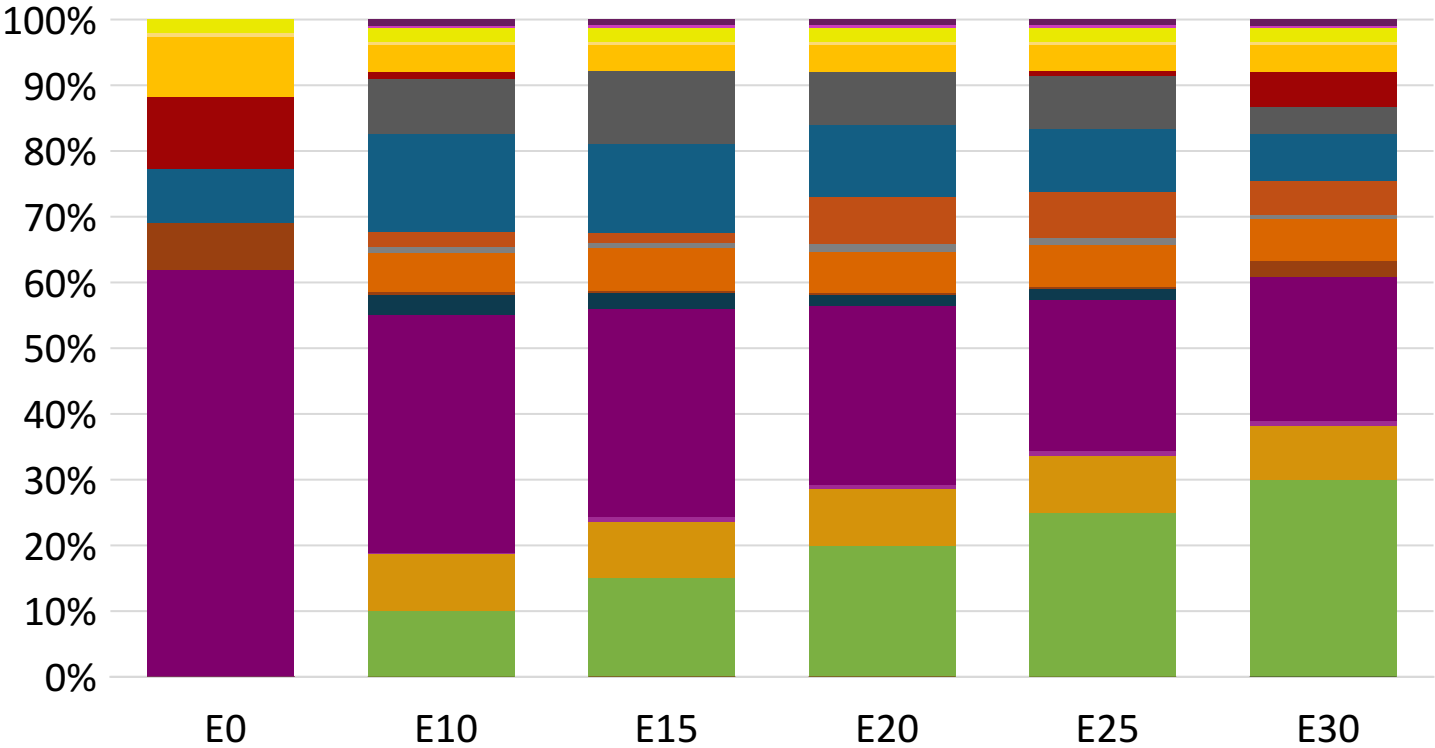


Average CIF 2024 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177

Source: Faro90

Italy – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

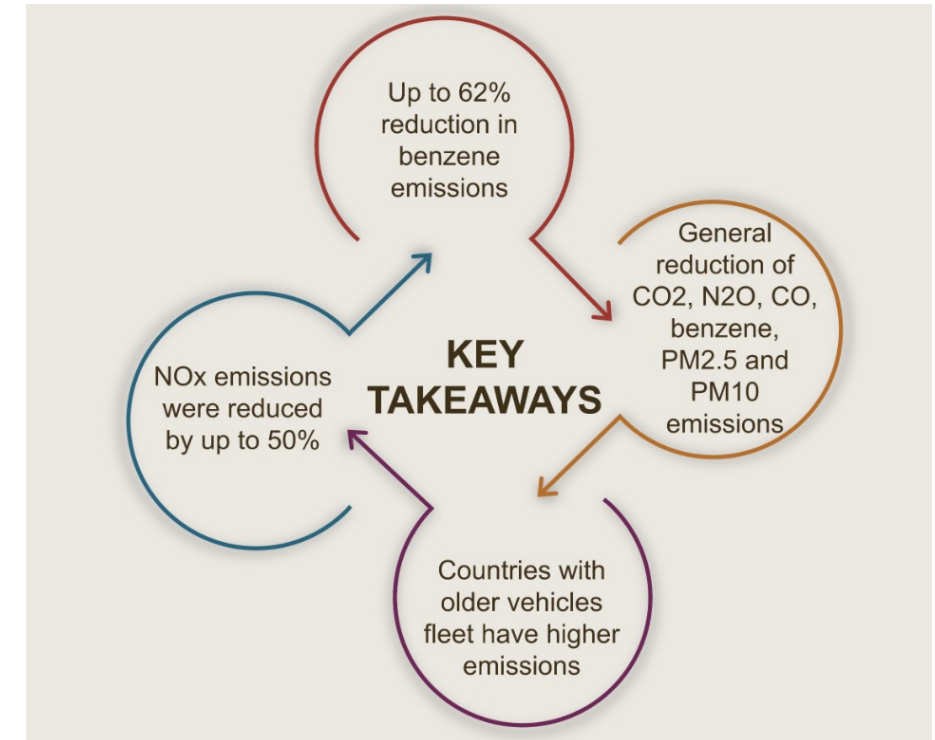
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

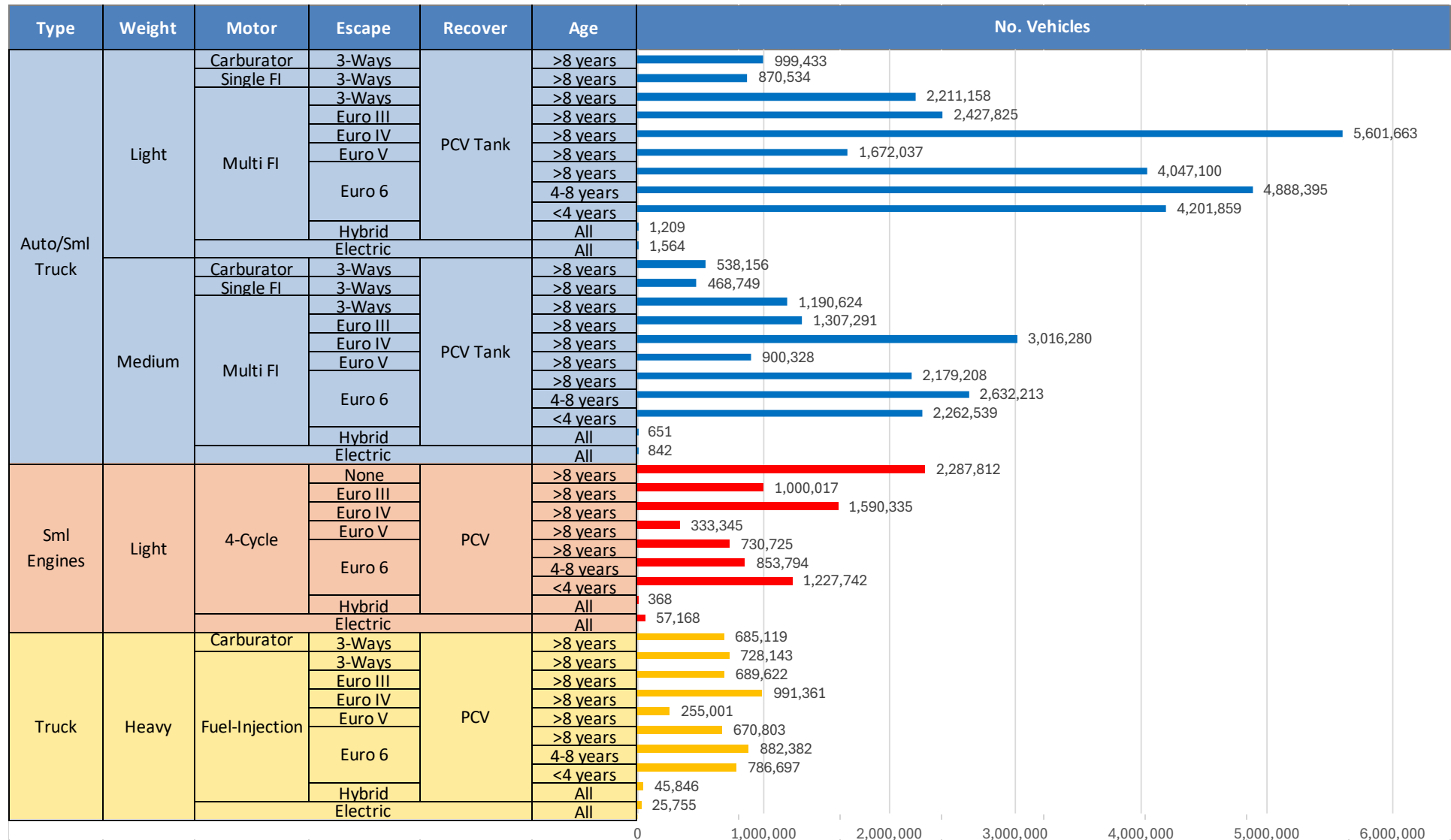
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Italy

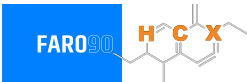


Vehicle Fleet: 55,261,692
Gasoline Vehicle Fleet: 19,966,496

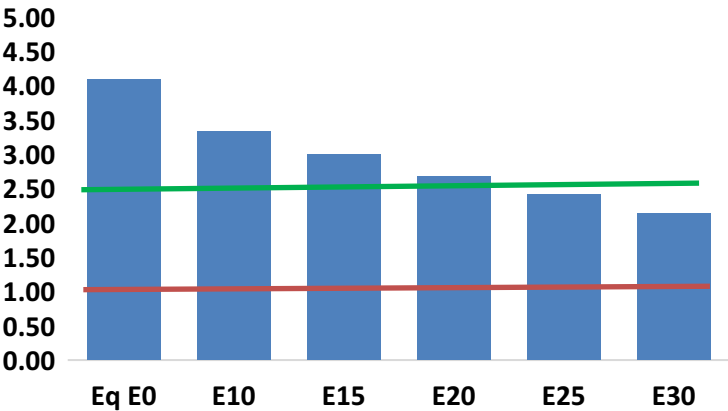
Source: Eurostat, ACEA

Italy – Vehicular Emissions

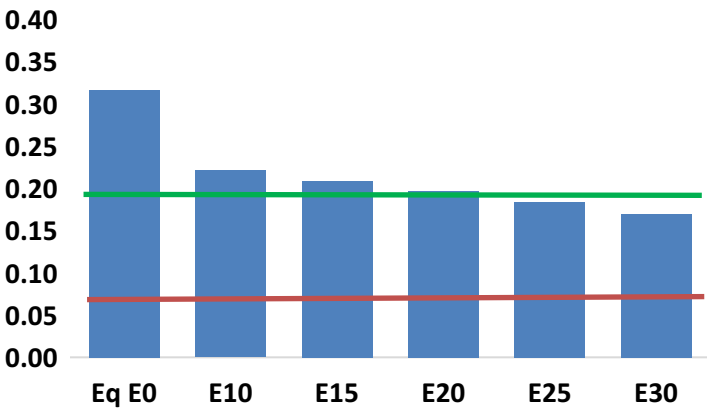
TIER III BIN 125 United States
Euro 6 Standard



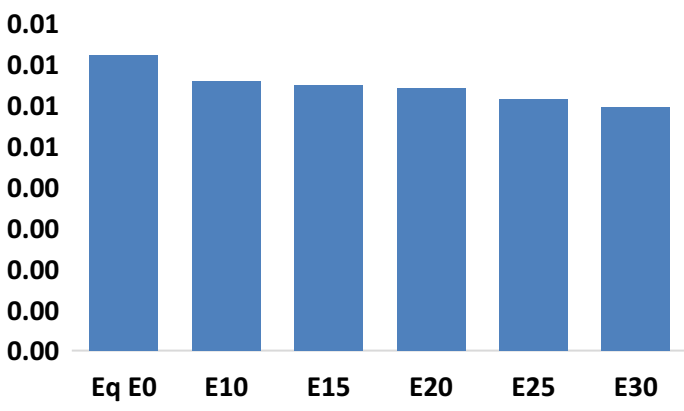
Carbon Monoxide (CO) g/km



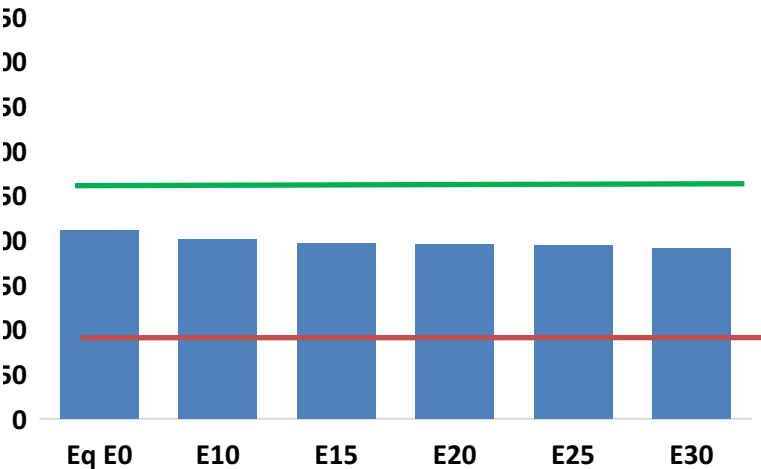
Nitrogen Oxides (NOx) g/km



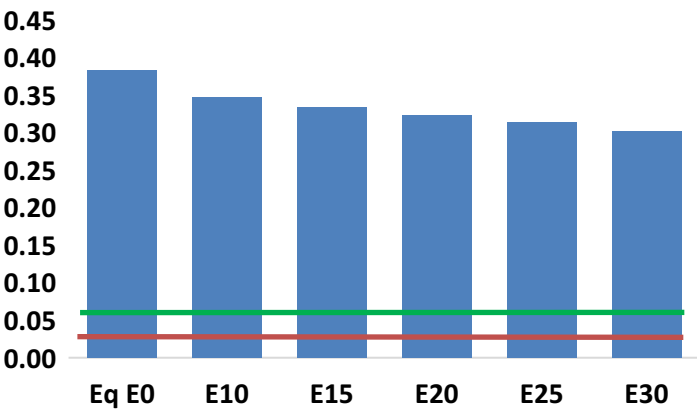
Aromatics (BTX) g/km



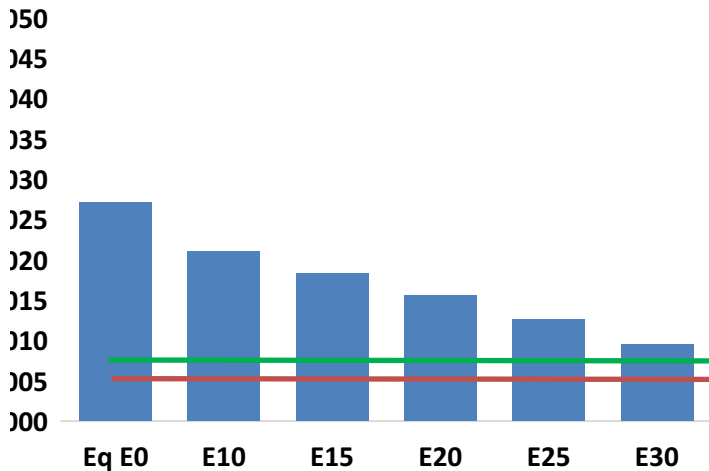
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



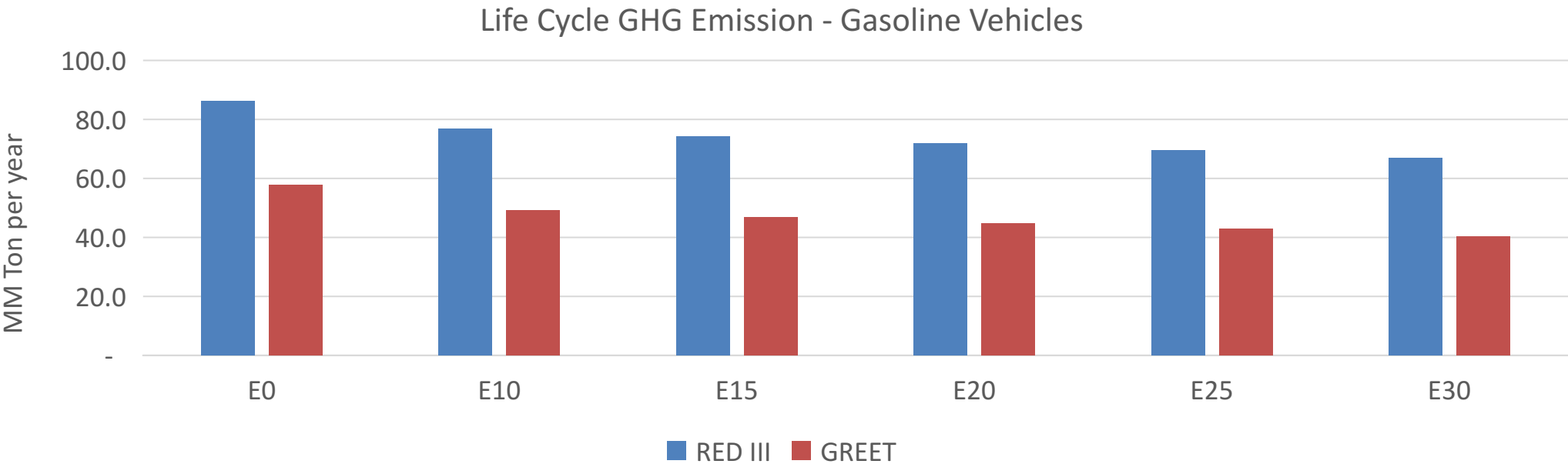
Particulate Matter (PM2.5) g/km



Italy – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,868,067	1,694,065	1,520,062	1,371,545	1,219,358	1,101,511	973,684	-19%	-35%
CO2	96,686,516	94,200,291	91,852,190	90,005,478	89,091,652	88,644,245	87,010,277	-5%	-8%
VOC	175,091	166,801	158,512	152,728	147,372	143,394	137,590	-9%	-16%
NOx	144,409	108,307	101,087	95,274	89,852	83,873	77,548	-30%	-38%
SOx	1,112	1,028	944	873	805	731	653	-15%	-28%
NH3	33,122	32,794	32,466	32,620	32,987	33,245	33,460	-2%	0%
N2O	1,578	1,570	1,563	1,571	1,603	1,621	1,640	-1%	2%
CH4	40,511	40,511	40,511	41,118	42,010	42,779	43,509	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,540	1,462	1,384	1,328	1,275	1,236	1,179	-10%	-17%
Acetaldehyde	3,291	3,777	5,542	8,440	11,498	13,139	15,525	68%	249%
Formaldehyde	10,962	10,658	12,424	14,588	15,283	16,907	18,406	13%	39%
Benzene	3,310	3,181	3,014	2,967	2,942	2,814	2,723	-9%	-11%
PM2.5	5,909	5,246	4,584	3,981	3,381	2,743	2,079	-22%	-43%
PM10	6,477	5,751	5,025	4,364	3,706	3,007	2,279	-22%	-43%

Italy – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	357.1
Target @ 2035 (MMT/year)	201.0
Delta	156.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.7%	18.8%	22.3%	25.7%	30.0%
RED II/III	0.0%	10.8%	14.0%	16.7%	19.3%	22.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	7.0%	8.3%	9.5%	11.1%
RED II/III	0.0%	6.0%	7.7%	9.2%	10.6%	12.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90